

Mathematical modeling of drone applications in disaster scenarios

Theorem or Problem: The lack of rapid and efficient wildfire response, which could be resolved through the implementation of drone technology.

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Summary

This project aims to optimize the deployment and operation of drones in natural disaster response using mathematical modeling. By developing probabilistic models and advanced algorithms, we seek to enhance the efficiency of drone-based supply delivery, real-time data collection, and rescue operations. Our approach incorporates key mathematical concepts, including geometric means, differential equations, and vector mathematics, to improve drone performance and reliability. Additionally, we account for real-world factors such as wind conditions and situational constraints to ensure practical and effective optimization. The research will be conducted in accordance with OCSEF guidelines and meticulously documented in a detailed Lab Notebook.

Background Research

Drones have been used previously in disaster relief efforts such as during the 2017 Hurricane Harvey, or the 2020 California Wildfires. Drones have been used to aid rescue efforts by assessing damage and delivering small items for immediate relief. Scientific papers such as *Drone Applications for Emergency and Urgent Care: A Systematic Review* state that “One of the main benefits of drones is to avoid endangering rescuers in shootings, fires, radiation, the presence of infectious agents, explosives, smoke, or gases.” A drone possesses the advantage of small size, allowing it to reach areas rescuers would normally not be able to.

Although drones have been used in previous disaster relief efforts, there have been no systematic approaches to the efficient deployment of drones to minimize damage to life and property. We seek to remedy that, and establish an efficient grid of drone technology to respond as quickly as possible and reduce damage. The recent Palisades and Eaton Fire is proof of this; the lack of rapid response contributed to the already disastrous effects of the fires. By utilizing new technologies like AI modeling, LiDAR, and UAS, drones can be heightened to new levels of efficiency and collect important information to fight fires and other natural disasters.

Theorem or Problem

Research question: How can mathematical modeling optimize the deployment of drones for disaster response?

- *We aim to develop algorithms and sets of equations that will aid in finding the the most effective system in using drones for disaster response*
- *We aim to make a system that can calculate the number of potential hazard zones that each drone can cover, incorporating factors such as air resistance and battery life.*
- *We will then determine optimal drone launch center locations using geometric median analysis, accounting for wind conditions.*
- *This entire process will be determining hypothetical system Irvine can use for drone deployment. This is a simulation of our research question through mathematical modeling, allowing us to test different scenarios and improve deployment strategies for disaster response. From this we can refine our models to ensure efficient and effective drone operations*

Notation and Framework

We will be calculating the geometric mean location of possible drone launch center. After We are using the QGIS(geographic information system software)

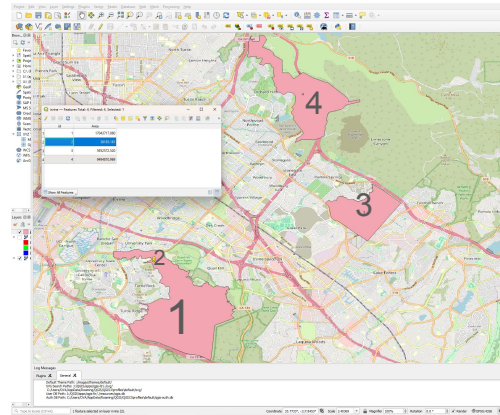
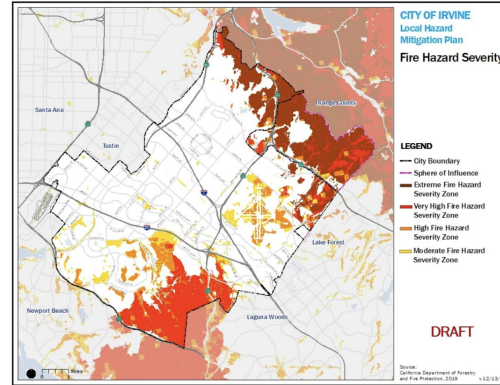
Used the Orange County Fire Authority Fire Hazard Severity Zones map from 2012. Georeferenced the image using the WGS 84 coordinate reference system (EPSG:4326), digitized the areas into polygons, and calculated each field's area in square meters.

Coordinate

1. 33.62057,-117.79181 (area: 17043717.860)
2. 33.654105,-117.807299 (area: 38185.161)
3. 33.67728,-117.69881 (area: 5952572.520)
4. 33.72948,-117.72014 (area: 9494810.969)

Average annual wind speed: 8 miles per hour
In West direction (assuming spring season)

Another used fact is that there is there is estimated 6.5C decrease in temperature with every 1000m of altitude increase (in small heights)



Thermal efficiency model
$$T_b = \frac{C \cdot \eta}{P_{total}}$$

T_b = Battery lifetime
 C = Battery capacity
 η = Battery efficiency factor (varies with temperature),
 P_{total} = Total power consumption.

Temperature Effect (η)
$$\eta(T) = 1 - k(T_{nominal} - T)$$

 $k = 0.01$ (empirical constant in small temperature deviations),
 $T_{nominal} = 15C(288K)$.
 T = temperature in Kelvin.

Results – Data/Observations (cont'd)

More accurate time and distance calculation with wind and realistic trajectory.

We need to take ascension with the wind into account. Drone needs to reach vertical distance of k meters when drone can ascend 6 m/s and horizontal wind speed of w m/s

$\frac{k}{6}$ secs with horizontal distance of $\frac{k \cdot w}{6}$ m

Let the point where the drone starts traveling be $m = (i, j)$ where the distance from the origin to m is $kw/6$ meters.

Let vector u be the vector from m to the destination

Let vector w and d be the vectors for the wind and drone. Then $\theta_1 = \arccos(\frac{w \cdot u}{|w||u|})$

$$\theta_2 = \arcsin(\frac{|w| \sin(\theta_1)}{|d|})$$

$$x = \frac{\sin(180^\circ - \theta_1 - \theta_2) |d|}{\sin(\theta_1)}$$

Where x is the actual drone's speed.

This will account for wind push in the air.

The accurate distance formula will be

$$d = 195.1 + \frac{kw}{6} + x(t - 11.5663 - \frac{k}{6}) \text{ for } t \geq 11.5663 + \frac{k}{6}$$

i.e. let's say that we need to ascend 1000m and the wind is blowing 4.5 m/s SW. The destination is 3000 m west and 4000 m south from the starting point.

166.67 seconds with the distance of 750 m traveled.

$$m = (-1500/\sqrt{2}, -1500/\sqrt{2}).$$

$$u = (-3000 + 1500/\sqrt{2}, -4000 + 1500/\sqrt{2})$$

$$x = 27.3906$$

So the drone will travel at the speed of 27.3906 m/s where the distance equation looks like

$$d = 195.1 + 750 + 27.3906(t - 178.2363) \text{ for } t \geq 178.2363$$

Notation and Framework (cont'd)

Drag Force (Air Resistance): The force opposing a drone's motion due to air, given by:

$$F_{\text{drag}} = 1/2 \rho C A v^2$$

$$F = m \cdot a$$

$$m \cdot a_{\text{drag}} = 1/2 \rho C A v^2, \text{ where}$$

ρ = air density (1.293 kg m⁻³)

C = drag coefficient (0.5117)

A = cross sectional area (0.5427 m²)

v = velocity

M = mass of the drone (3.6 kg)

This drag force will be plugged into an equation to generate a

final differential equation,

which is then solved and

utilized throughout the results.

$$\frac{dv}{dt} = -0.00919v^2 + 5;$$

Many constants are from the Support for Matrice 300 RTK, or derived, which are in the Results slide.

The geometric median minimizes the sum of distances to a set of given points, crucial for determining optimal drone launch sites.

Total distance will be calculate with

$$x^* = \arg \sum_{i=1}^n ||X - X_i||$$

Where:

$||X - x_i||$ is the euclidean distance between the candidate location x and hazard zone x_i

In this research, the flight duration for all drones is kept constant. Given uniform wind speed and direction, each drone will experience identical displacement during flight. Therefore, the required procedure involves using the battery drainage time to determine the displacement caused by the wind using the equation $d=vt$. Since this displacement will be the same for all drones, the launch site should be shifted uniformly in the opposite direction by the same magnitude to compensate accordingly.

Velocity of the drone with the account of air resistance

$$v(t) = v_d \tanh(av_d t)$$

Where v_d is maximum velocity of the drone and a is the acceleration of the drone

Position of the drone (how far away it is from point of departure)

$$s(t) = \frac{\ln(\cosh(av_d t))}{a}$$

Results – Data/Observations

Solving out the differential equation $\frac{dv}{dt} = -0.00919v^2 + 5$, we get that $v = \sqrt{\frac{5}{r}} \tanh(5t * \sqrt{\frac{r}{5}})$, where $r = 0.00919$. Taking the limit of the velocity function as it goes to infinity, $v = \lim_{t \rightarrow \infty} \sqrt{\frac{5}{r}} \tanh(5t * \sqrt{\frac{r}{5}}) = 23.3253004464$ which is actually very close to the maximum velocity of 23m/s that the user's manual provides, verifying that the calculated model is accurate. Assuming that 23m/s is the maximum velocity to make computation easier, we get that for the displacement of the drone is $d = 195.1 + 23(t - 11.5663)$ for $t \geq 11.5663$ (this is derived after calculating that it will take 11.5663s to reach the maximum velocity, and then taking the integral $\int_0^{11.5663} (\sqrt{\frac{5}{r}} \tanh(5t * \sqrt{\frac{r}{5}}) dt = 195.1m$ to find the displacement while speeding up). Given that the battery life of drone is 55 minutes, the maximum travel distance of the drone is 75.829 km, giving that the area of coverage is 18,064.2747 km². While the size of Irvine is small with an area of 170.74 km², when applied to a larger region, this process of calculating the travel distance and its results may be significant when there has to be more than one drone station. This is because knowing the area of coverage per each station/drone will be used to calculate the number of drone stations. For example, if implementing this in the entire state of California, there should be 423,789 km² (area of California) / 18,064.2747 km² $\approx$ stations. (This is an underestimate because the area of coverage per each station is circular).

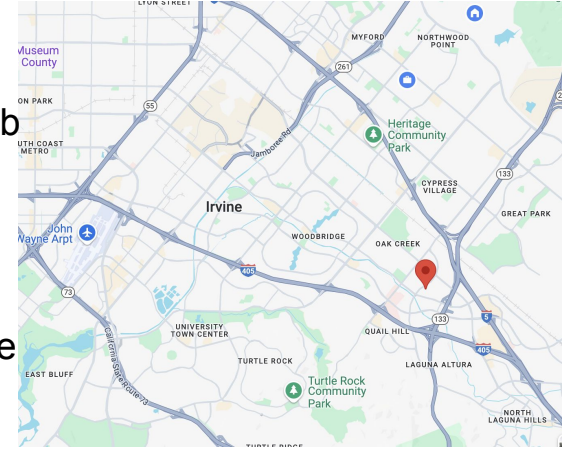
Results – Data/Observations (cont'd)

Battery calculation with temperature:

Battery performance model: $T_b = \frac{C \cdot \eta}{P_{total}}$
Because we are only concerned about battery efficiency factor, we get that

$T_b = T_{ideal} (1 - k(T_{nominal} - T))$ given that
. Assuming that the drone goes up to 1000m altitude, the temperature would be around 8.5C(281.5K). T_{ideal} is 55 minutes for the drone. Plugging in all the above constants, we get that the battery life at the cruising altitude is 51.425 minutes, which is very close to the value of 55 minutes that we were using. Knowing that there are inherent protective systems within batteries to minimize the performance drop, we know that temperature only affects the battery performance in a negligible amount in this scenario.

The computed geometric median was (33.66344, -117.7635). The simulation code, provided in the Lab Notebook screenshots, iterates through fire-hazard areas, each represented by a centroid calculated using QGIS. The total linear travel distance for each drone is computed and weighted by the area of each region. The point minimizing total travel distance is selected. The program, built with NumPy, is highly scalable, allowing easy updates to points and areas. In order to account for the wind effects, we calculate the expected duration of the flight using vectors. To area 1: 27.4m/s, 199 secs; area 2: 35.5m/s, 118 secs; Area 3: 10.3 m/s, 596 secs; Area 4: 13.3m/s, 628 secs. While the geometric median ensures the shortest avg. distance (6040.5m), it is better to shift the drone station around 4761 m to the East since (weighted) average time is 454s. If we move the station 4761 meters to the East, the avg time spent traveling will be reduced to 388 seconds.



Discussion and Conclusion

Limitations: Although the mathematical expressions that we utilized throughout our model are proven equations, they might still have flaws and could have inconsistent results. As a result, our findings should not be considered immediately conclusive but rather as an initial approximation of how drone deployment strategies could be applied in real-world disaster scenarios. Further validation through empirical testing and refinement of the model is necessary for practical implementation. However, because of the fact that the drone launch centers may

Challenges: Our project involves establishment of facilities that support drone technology, which would be costly to build and maintain. Another challenge was trying to find geometric median, which we had to find a compromise to use a computer simulation and only to apply to effect of wind after the calculation of the median, which may result in imperfect placement of the station.

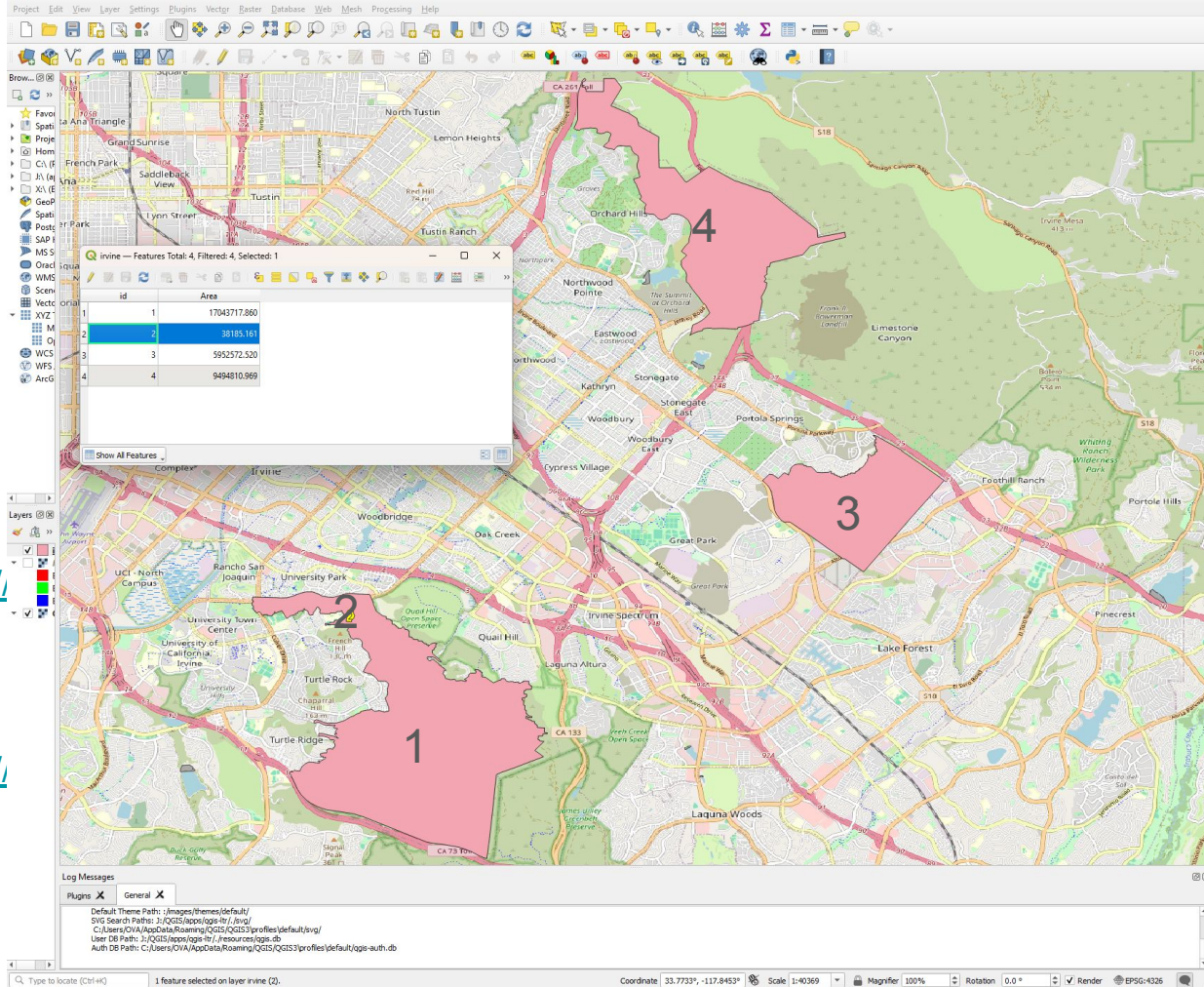
Applications: Our project could be the basis for the establishment of a system geared towards disaster risk mitigation and reducing the destructive impacts of catastrophes like the Palisades and Eaton Fires. Even if cost might be high, the monetary benefits from blocking further damage would far outweigh the initial cost of the development of these facilities.

Reflection/Application

40,000 acres, 18,000 structures, and 85 lives - this is what we lost in the Palisades and Eaton Fires. These aren't just numbers, they are lives, careers, and communities scarred forever by this disaster. In the face of such a formidable adversary, we must rise up and use technology to improve reaction times in an effort to reduce further losses. Our project on drone optimization creates a system where drones can aid first responders act faster and more efficiently. Optimized drone systems means that search and rescue teams are able to cover more ground and save lives in all types of disasters where a few minutes might mean the difference between life and death. Beyond disaster relief, these optimization techniques can be used in other fields such as wildlife monitoring and delivery logistics. By refining drone route optimization, we aren't just enhancing technology—we're safeguarding lives, livelihoods, and entire communities. First responders, emergency planners, and humanitarian organizations could all benefit from these advancements, ensuring faster and more effective disaster response. In the end, it's not just about efficiency, it's about the lives and homes we can save through this technology.

References Cited

- DJI. (2024). *Support for Matrice 300 RTK - DJI*. DJI.
<https://www.dji.com/support/product/matrice-300>
- Drones for Assessment of Disaster Damage and Impact - Revolutionizing Disaster Response | United Nations Development Programme*. (n.d.). UNDP.
<https://www.undp.org/vietnam/blog/drones-assessment-disaster-damage-and-impact-revolutionizing-disaster-response>
- How Drones are Used in the Aftermath of Disasters*. (2024, March 19). Amprius Technologies. <https://amprius.com/drones-disasters/>
- Ishiwatari, M. (2024). Leveraging Drones for Effective Disaster Management: A Comprehensive Analysis of the 2024 Noto Peninsula Earthquake Case in Japan. *Progress in Disaster Science*, 23, 100348.
<https://doi.org/10.1016/j.pdisas.2024.100348>
- Sanz-Martos, S., López-Franco, M. D., Álvarez-García, C., Granero-Moya, N., López-Hens, J. M., Cámara-Angueta, S., Pancorbo-Hidalgo, P. L., & Comino-Sanz, I. M. (2022). Drone Applications for Emergency and Urgent Care: A Systematic Review. *Prehospital and Disaster Medicine*, 37(4), 502–508.
<https://doi.org/10.1017/s1049023x22000887>



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